

Balanced ternary, transcendentals, and the geometry of number systems

****These five mathematical threads — balanced ternary arithmetic, fractional part digit extraction, transcendental constants, higher-dimensional geometry, and novel computing models — share genuine pairwise connections but do not form a single unified framework.**** The strongest links run between balanced ternary, modular arithmetic, and fractional part functions, where the algebra is tight and the identities are exact. The connections to π and higher-dimensional geometry are real but thinner: a proven impossibility result blocks base-3 BBP formulas for π itself (though π^2 has one), and Fermat's Last Theorem connects to ternary only through the Eisenstein integers $\mathbb{Z}[\omega]$. Meanwhile, the emerging qutrit quantum computing program provides the most promising bridge for future unification — it inherits the algebraic elegance of ternary arithmetic and embeds it in a richer Hilbert space geometry where L_p norms, lattice structures, and transcendental phases all naturally converge. This report lays out the rigorous mathematics of each domain and maps exactly where the connections hold and where they break.

The algebraic elegance of balanced ternary is real and provable

A number N in balanced ternary is expressed as **$N = \sum_i d_i \cdot 3^i$** where each digit $d_i \in \{-1, 0, 1\}$. [Wikipedia] (https://en.wikipedia.org/wiki/Balanced_ternary) Every integer has a unique canonical representation (no leading zeros), and the sign of N is simply its leading nonzero digit — no sign bit, no two's complement. [Wikipedia] (https://en.wikipedia.org/wiki/Balanced_ternary) Negation is trivial: swap every $-1 \leftrightarrow 1$ and leave 0s unchanged. [University of Iowa +2] (<https://homepage.cs.uiowa.edu/~jones/ternary/arith.shtml>) This alone eliminates an entire class of hardware complexity that binary systems must manage. [Louis-dr] (<https://louis-dr.github.io/ternalu3.html>) [Grokopedia] (https://grokopedia.com/page/Balanced_ternary)

The radix economy argument establishes base 3 as optimal among integers. The cost of representing N in base b is approximately **$E(b, N) \approx b \cdot \ln(N)/\ln(b)$** , so minimizing $b/\ln(b)$ yields $b = e \approx 2.718$ [Wikipedia] (https://en.wikipedia.org/wiki/Radix_economy) [williams] (https://web.williams.edu/Mathematics/sjmiller/public_html/105Sp10/addcomments/Hayes_ThirdBase.htm) (Steiner's problem, 1850). Evaluating at integer bases: **$b = 3$ gives $3/\ln 3 \approx 2.731$** , only 0.46% worse than the theoretical optimum, while $b = 2$ and $b = 4$ both give $2/\ln 2 \approx 2.885$, some 6.15% worse. [Williams College] (https://web.williams.edu/Mathematics/sjmiller/public_html/105Sp10/addcomments/Hayes_ThirdBase.htm) The original engineering analysis appeared in *High-Speed Computing Devices* (ERA, 1950), which computed triode counts for registers covering numbers to 10^6 : radix 3 required **38.24 triodes** versus binary's 39.20 [williams] (https://web.williams.edu/Mathematics/sjmiller/public_html/105Sp10/addcomments/Hayes_ThirdBase.htm) [Williams College] (https://web.williams.edu/Mathematics/sjmiller/public_html/105Sp10/addcomments/Hayes_ThirdBase.htm) — a modest but mathematically real advantage. [Williams College] (https://web.williams.edu/Mathematics/sjmiller/public_html/105Sp10/addcomments/Hayes_ThirdBase.htm)

The arithmetic tables reveal deeper structure. The single-trit multiplication table is **entirely carry-free**: multiplying by -1 is negation, by 0 is zero, by 1 is identity. The addition table has only **2 carries out of 9 entries** [Wikipedia] (https://en.wikipedia.org/wiki/Balanced_ternary) (when $1 + 1 = 3 \equiv -1$, producing digit -1 with carry $+1$, and symmetrically for $-1 + -1$). [Wikipedia] (https://en.wikipedia.org/wiki/Balanced_ternary) Donald Knuth's observation that **truncation equals rounding** [Wikipedia +2] (https://en.wikipedia.org/wiki/Balanced_ternary) follows from the fact that the maximum fractional value is $\sum_{k=1}^{\infty} 1 \cdot 3^{-k} = 1/2$, so the fractional part always lies in $[-1/2, 1/2]$, meaning truncation automatically selects the

nearest integer. [Wikipedia](https://en.wikipedia.org/wiki/Balanced_ternary) [johndcook]
(https://www.johndcook.com/blog/2025/01/23/balanced-ternary-rounding/)

The Setun computer (1958, Moscow State University, designed by Nikolai Brusentsov) [Earl T. Campbell]
(https://earlrcampbell.com/2014/12/29/the-setun-computer/) [Grokopedia]
(https://grokopedia.com/page/Balanced_ternary) demonstrated these properties in hardware. [Wikipedia]
(https://en.wikipedia.org/wiki/Ternary_computer) [Wikipedia](https://en.wikipedia.org/wiki/Setun) It used 18-trit
words [ScienceDirect](https://www.sciencedirect.com/science/article/abs/pii/S0026269218304634) stored in
paired ferrite cores, executed addition in 180 μ s and multiplication in 335 μ s, and ran asynchronously without a
central clock. [George Mason University](https://mason.gmu.edu/~drine/History-of-Ternary-Computers.htm)
About 50 units were built; [Earl T. Campbell](https://earlrcampbell.com/2014/12/29/the-setun-computer/) when
replaced by binary machines, the replacements cost **2.5 $\times$ more** for equivalent performance. [Wikipedia]
(https://en.wikipedia.org/wiki/Setun) The successor Setun-70 implemented Dijkstra's structured programming in
hardware, using 6-trit syllables [Wikipedia](https://en.wikipedia.org/wiki/Setun) and a two-stack architecture
[Wikipedia](https://en.wikipedia.org/wiki/Ternary_computer) that Brusentsov claimed reduced programming
time by 5–10 $\times$. [Wikipedia](https://en.wikipedia.org/wiki/Setun)

Modern ternary computing research has accelerated sharply. Samsung funded a team at UNIST that published
the first energy-efficient ternary metal-oxide semiconductor on an 8-inch wafer [Wikipedia]
(https://en.wikipedia.org/wiki/Ternary_computer) in *Nature Electronics* (2019). Huawei filed patent
CN119652311A (published March 2025) describing ternary logic gate circuits [South China Morning Post]
(https://www.scmp.com/tech/big-tech/article/3305201/tech-war-huaweis-ternary-logic-patent-could-solve-
problem-power-hungry-ai-chips) claiming 40% transistor count reduction. [Frank's World]
(https://www.franksworld.com/2025/09/24/ternary-computing-breakthrough-explained/) [Redirect]
(https://min.news/en/tech/67b88b05ab16b159ea4f78e36fe0fa98.html) In January 2025, *Science Advances*
published high-performance ternary logic circuits using carbon nanotube source-gating transistors, [PubMed
Central](https://pmc.ncbi.nlm.nih.gov/articles/PMC11721562/) [ResearchGate]
(https://www.researchgate.net/publication/377783663_Carry-
Free_Adder_in_Redundant_Number_System_the_First_and_Only_Demonstration_of_Ternary_Logic's_Superiority_Over_Binary
and *Nature Communications* reported reconfigurable binary/ternary floating-gate heterojunction devices.
[Nature](https://www.nature.com/articles/s41467-025-62116-y.pdf) Perhaps most strikingly, Microsoft
Research's BitNet b1.58 demonstrated that large language models with **ternary weights $\{-1, 0, +1\}$** —
encoding $\log_2(3) \approx 1.58$ bits per weight — can match full-precision model accuracy while cutting energy use up to
3 $\times$. [Aman Chourasia](https://www.amanchourasia.in/2025/09/ternary-computing-in-artificial-intelligence-chip-
design.html)

Fractional parts and the centering shift that defines balanced ternary

The fractional part function $\{x\} = x - \lfloor x \rfloor$ is the engine of digit extraction in any base. [Grokopedia]
(https://grokopedia.com/page/Fractional_part) For base b , the k -th digit is **$d_k = \lfloor b \cdot \{b^{k-1} \cdot x\} \rfloor$** , and the
iteration $T(x) = \{bx\}$ (the "times- b map") shifts the base- b expansion one position left. For base 3, this becomes
the **tripling map** $T(x) = 3x \bmod 1$, fundamental in ergodic theory and central to the construction of the Cantor
set. [Cut the Knot](https://www.cut-the-knot.org/blue/frac_conv.shtml)

Converting from standard base-3 digits $\{0, 1, 2\}$ to balanced ternary digits $\{-1, 0, 1\}$ requires what is essentially a
centering operation. The formal algorithm repeatedly computes $r = n \bmod 3$; when $r = 2$, it substitutes $r = -1$ and

adds a carry of +1 to the next position. [GeeksforGeeks](https://www.geeksforgeeks.org/dsa/balanced-ternary-number-system/) This is the **symmetric residue system mod 3**, defined by $\text{sym_mod}(n, 3) = n - 3 \cdot \text{round}(n/3)$.

The expression $\{x + 1\} - 1/3 = \{x\} - 1/3$ (since $\{x + 1\} = \{x\}$ by periodicity) [Grokopedia](https://grokopedia.com/page/Fractional_part) arises naturally as the per-digit rebalancing shift. Consider what $\{x\} - 1/3$ does to the three standard ternary digit intervals: it maps $[0, 1/3) \rightarrow [-1/3, 0)$, maps $[1/3, 2/3) \rightarrow [0, 1/3)$, and maps $[2/3, 1) \rightarrow [1/3, 2/3)$. These intervals correspond precisely to balanced ternary digits $-1, 0$, and $+1$ respectively, each centered at the digit value scaled by $1/3$. The total cumulative bias for converting an unsigned fractional representation to balanced ternary is $\sum_{k=1}^{\infty} 3^{-k} = 1/2$, and $1/3$ is the first-digit contribution to this shift — the per-digit piece of the centering that makes balanced ternary work.

The fractional part function connects to base-3 through several exact identities. Hermite's identity specialized to $n = 3$ gives:

$$\{x\} + \{x + 1/3\} + \{x + 2/3\} = \{3x\} + 1$$

This remarkable formula says the sum of fractional parts at three equally-spaced shifts equals the fractional part of the tripled argument plus 1. It is periodic with period $1/3$ (shifting x by $1/3$ cyclically permutes the three terms), and it directly encodes the ternary structure of the real line.

The Cantor set provides another deep base-3 connection. [Wikipedia](https://en.wikipedia.org/wiki/Ternary_numeral_system) Defined as the set of $x \in [0,1]$ whose base-3 expansion uses only digits $\{0, 2\}$, [Brilliant](https://brilliant.org/wiki/cantor-set/) it is characterized by the condition $\lfloor 3^k \cdot x \rfloor \neq 1$ for all k . [Wikipedia](https://en.wikipedia.org/wiki/Ternary_numeral_system) [williams](https://web.williams.edu/Mathematics/sjmiller/public_html/105Sp10/addcomments/Hayes_ThirdBase.htm) The Cantor function (Devil's Staircase) converts base-3 to base-2 by replacing each 2 with 1 [Blacksacademy](https://blacksacademy.net/Poincareweb/063%20The%20Cantor%20set.htm) — a continuous, monotone surjection from $[0,1]$ to $[0,1]$ with derivative zero almost everywhere. The iterated function system $G_0(x) = x/3$, $G_2(x) = (x+2)/3$ generates the Cantor set, and the 3-adic integers $\mathbb{Z}_3$ contain a natural homeomorphic copy [Cornell Math](https://pi.math.cornell.edu/~fractals/6/slides/Lu.pdf) with Hausdorff dimension $\log_3 2$.

π resists base-3 digit extraction — but π^2 does not

The relationship between π and base-3 is marked by a **proven impossibility**. Borwein, Galway, and Borwein (2004) showed that no degree-1 BBP-type formula for π exists except in bases that are powers of 2. [davidhbailey](https://www.davidhbailey.com/dhbpapers/bbp-formulas.pdf) The original BBP formula $\pi = \sum_{k=0}^{\infty} (1/16^k) [4/(8k+1) - 2/(8k+4) - 1/(8k+5) - 1/(8k+6)]$ extracts hexadecimal digits without computing preceding ones. [UNT Digital Library](https://digital.library.unt.edu/ark:/67531/metadc1013585/) [davidhbailey](https://www.davidhbailey.com/dhbpapers/bbp-formulas.pdf) In base 3, one can construct BBP-style series that produce combinations of π with arctangent and logarithm terms, but π is always **entangled with other constants** that cannot be isolated. [MetaFilter](https://ask.metafilter.com/360896/Math-formula-that-only-works-in-some-base-systems) A special cancellation in the polylogarithm at roots of unity occurs only for base-2 powers.

However, **π^2 has a base-3 BBP formula** [MetaFilter](https://ask.metafilter.com/360896/Math-formula-that-only-works-in-some-base-systems) [davidhbailey](https://www.davidhbailey.com/dhbpapers/bbp-formulas.pdf) (Formula 76 in Bailey's Compendium, updated April 2023). This is mathematically significant: the squaring

operation somehow "untangles" π from the obstructions that prevent a base-3 formula for π itself. Bailey's normality research suggests that under "Hypothesis A" (a conjecture about equidistribution of certain chaotic iterations), any constant with a BBP formula is either rational or normal to the corresponding base [Rug] (https://fse.studenttheses.ub.rug.nl/8432/1/Jurjen_Heegen_WB_2008.pdf) [davidhbailey] (<https://www.davidhbailey.com/dhbpapers/bbp-formulas.pdf>) — meaning π^2 may be provably normal in base 3, even though π 's normality in any base remains open.

The normality question for π in base 3 — whether all ternary digit strings appear with equal limiting frequency — is one of the major open problems in number theory. [Pi2e] (<https://pi2e.ch/blog/2016/10/27/is-pi-normal/>) Statistical analysis of trillions of digits shows behavior consistent with normality in every base tested, but no proof exists for any base. [Wikipedia] (<https://en.wikipedia.org/wiki/Pi>) The ternary expansion begins 10.010211012222010211002111110221222220... (OEIS A004602). [Wikipedia] (<https://en.wikipedia.org/wiki/Pi>) Borel's theorem guarantees that almost all real numbers are absolutely normal, but "almost all" is useless for specific constants. [Wikipedia] (https://en.wikipedia.org/wiki/Normal_number)

The deeper hierarchy here separates algebraic/transcendental from computable/non-computable. All algebraic numbers are computable, but the converse fails spectacularly: π is both transcendental and computable. [Wikipedia] (https://en.wikipedia.org/wiki/Computable_number) Chaitin's halting probability $^*\Omega = \sum_{\{p \text{ halts}\}} 2^{-|p|}$ is well-defined, transcendental, provably normal (being algorithmically random), but non-computable — knowing its first N bits would solve the halting problem for all programs of length $\leq N$. [Wikipedia] (https://en.wikipedia.org/wiki/Chaitin's_constant) The Hartmanis-Stearns conjecture (1965, still open) posits that no irrational algebraic number can have its base- b expansion computed in real time, [ResearchGate] (https://www.researchgate.net/publication/290527496_On_the_computational_complexity_of_algebraic_numbers_the_Hartmanis-Stearns_problem_revisited) [Rjlipton] (<https://rjlipton.com/2012/06/15/why-the-hartmanis-stearns-conjecture-is-still-open/>) establishing that computational speed separates rational, algebraic-irrational, and transcendental numbers more finely than algebraic classification alone.

The bridge between continuous constants and discrete computation runs primarily through the FFT (where π enters via the roots of unity $\omega = e^{2\pi i/n}$), Stirling's approximation ($n! \approx \sqrt{2\pi n}(n/e)^n$ pervades algorithm analysis), and spigot algorithms that stream digits of π using only integer arithmetic. [Cut the Knot] (<https://www.cut-the-knot.org/Curriculum/Algorithms/SpigotForPi.shtml>) The BBP algorithm itself is the sharpest bridge: it converts a transcendental constant into modular exponentiation — a purely discrete operation. [Wikipedia] (https://en.wikipedia.org/wiki/Bailey%E2%80%93Borwein%E2%80%93Plouffe_formula)

From Pythagoras to Fermat: geometry constrains what numbers can do

The Pythagorean theorem generalizes to N dimensions in two distinct ways. The L2 distance formula $d = \sqrt{\sum (x_i - y_i)^2}$ extends immediately, and de Gua's theorem (1783) establishes that for an n -simplex with a right-angle vertex, **the squared content of the hypotenuse face equals the sum of squared contents of the leg faces**. [En Academic] (<https://en-academic.com/dic.nsf/enwiki/3281113>) The proof uses the Cauchy-Binet formula: if A is the matrix of edge vectors from the right-angle vertex, then $\det(A^T A) = \sum_S [\det(A_S)]^2$, where the sum runs over all $k \times k$ submatrices. [Matt Bognar] (<https://homepage.divms.uiowa.edu/~frohman/pyth2.pdf>)

The L_p norms $\|x\|_p = (\sum |x_i|^p)^{1/p}$ parameterize a family of geometries. [Wikipedia] ([https://en.wikipedia.org/wiki/Norm_\(mathematics\)](https://en.wikipedia.org/wiki/Norm_(mathematics))) At $p = 1$ (Manhattan distance), the unit ball is a cross-polytope; at $p = 2$, a sphere; [University of Utah] (<https://users.cs.utah.edu/~jeffp/teaching/cs5955/L7-Distances.pdf>) [Fau] (<https://lme.tf.fau.de/lecture-notes/lecture-notes-pr/lecture-notes-in-pattern-recognition->

episode-17-norms-and-unit-balls/) at $p = 3$, a superellipse satisfying $|x|^3 + |y|^3 = 1$ with slightly flattened sides; at $p \rightarrow \infty$, a hypercube. [John D. Cook](https://www.johndcook.com/blog/2010/07/02/volumes-of-generalized-unit-balls/) The dual of L_p is L_q where $1/p + 1/q = 1$, connected through Hölder's inequality. These norms have direct computational applications: **L1 regularization (LASSO)** drives coefficients to exactly zero for sparse models, [Lunartech](https://www.lunartech.ai/blog/mastering-l1-and-l2-regularization-the-definitive-guide-to-preventing-overfitting-in-neural-networks) [ScienceDirect](https://www.sciencedirect.com/topics/engineering/lp-norm) **L2 regularization (Ridge)** shrinks uniformly, and the "L0 norm" (Hamming weight) counts nonzero entries [Oreate AI](https://www.oreateai.com/blog/detailed-explanation-of-norms-and-distance-measurement-methods/5dbc80fcab76b38407c7e882e5c4fb30) [Fau](https://lme.tf.fau.de/lecture-notes/lecture-notes-pr/lecture-notes-in-pattern-recognition-episode-17-norms-and-unit-balls/) — foundational for compressed sensing.

Fermat's Last Theorem provides the deepest geometric constraint relevant here. For $n = 2$, integer solutions to $a^2 + b^2 = c^2$ parametrize rational points on the unit circle — infinitely many exist. For **$n = 3$** , the equation $a^3 + b^3 = c^3$ having no positive integer solutions** means the Fermat curve $x^3 + y^3 = 1$ has no nontrivial rational points.

[Universiteit Leiden]

(https://www.universiteitleiden.nl/binaries/content/assets/science/mi/scripties/bachelor/2017-2018/viergever_bsc_juni-2018.pdf) Euler's original proof (1770) used infinite descent in $\mathbb{Z}[\sqrt{-3}]$, though it required repair [MacTutor History of Mathematics](https://mathshistory.st-andrews.ac.uk/HistTopics/Fermat's_last_theorem/) using the **Eisenstein integers** $\mathbb{Z}[\omega]$ where $\omega = e^{2\pi i/3}$ — a hexagonal lattice in the complex plane where unique factorization holds. The factorization $x^3 + y^3 = (x + y)(x + \omega y)(x + \omega^2 y)$ in $\mathbb{Z}[\omega]$, with the Eisenstein prime $\lambda = 1 - \omega$ satisfying $N(\lambda) = 3$, is the key step. [PRIMES](https://www.wstein.org/edu/2010/414/projects/ohana.pdf) [UCI Mathematics](https://www.math.uci.edu/~ndonalds/math180b/6gaussian.pdf)

This is where ternary arithmetic and higher geometry genuinely intersect. The Eisenstein integers form the natural algebraic setting for cubic reciprocity and FLT for $n = 3$, and they have norm $N(a + b\omega) = a^2 - ab + b^2$ — a quadratic form intimately connected to the geometry of the hexagonal lattice. [Emergent Mind]

(https://www.emergentmind.com/topics/eisenstein-integers) The prime 3 is special here: it ramifies as $3 = -\omega^2(1 - \omega)^2$, [Wikipedia](https://en.wikipedia.org/wiki/Gaussian_integer) making the Eisenstein prime $\lambda = 1 - \omega$ the fundamental building block. [arXiv](https://arxiv.org/abs/1602.06726) This is a genuine mathematical connection between base-3 structure and deep number theory, though it operates at the level of algebraic number theory rather than positional notation.

The computational geometry connection runs through lattice problems. The **Shortest Vector Problem (SVP)** and [NYU Courant](https://cims.nyu.edu/~regev/papers/pqc.pdf) **Learning with Errors (LWE)** — foundations of post-quantum cryptography [Medium](https://medium.com/@ankitacode11/post-quantum-cryptography-understanding-lattices-and-modern-cryptographic-problems-edd76721118c) (NIST standardized Kyber and Dilithium in 2024) — depend critically on the norm used. Under L_2 , SVP admits polynomial-time approximation via LLL for large approximation factors; under L_∞ , exact SVP is NP-hard. [Wikipedia](https://en.wikipedia.org/wiki/Lattice_problem) The balanced ternary lattice $\{-1, 0, 1\}^n$ is a natural search space for short vectors, and Generalized Balanced Ternary extends this to hexagonal tilings in higher dimensions.

Qutrits and the emerging convergence point

The most promising site for genuine unification of these threads is **qutrit quantum computing**. A qutrit lives in $\mathbb{C}^3$ with basis states $|0\rangle, |1\rangle, |2\rangle$ and general state $\alpha|0\rangle + \beta|1\rangle + \gamma|2\rangle$ [Wikipedia]

(https://en.wikipedia.org/wiki/Ternary_computer) where $|α|^2 + |β|^2 + |γ|^2 = 1$. [ScienceDirect] (<https://www.sciencedirect.com/science/article/abs/pii/S221053792300063X>) [Wikipedia] (<https://en.wikipedia.org/wiki/Qutrit>) Single-qutrit gates are 3×3 unitary matrices from $U(3)$, parameterized by 8 generators** [Wikipedia](<https://en.wikipedia.org/wiki/Qutrit>) (the Gell-Mann matrices of $su(3)$, the same Lie algebra governing the strong nuclear force). The qutrit quantum Fourier transform uses phases $e^{2\pi i/3^n}$ – embedding π directly into base-3 arithmetic.

Recent experimental progress has been rapid. Superconducting qutrits now achieve high-fidelity entangling gates [Nature](<https://www.nature.com/articles/s41534-024-00892-z>) (Goss et al., *Nature Communications*, 2022), and in 2024, *npj Quantum Information* published the first error mitigation experiments on qutrits. [Nature] (<https://www.nature.com/articles/s41534-024-00892-z>) Gokhale et al. (ISCA 2019) showed that qutrits can decompose generalized CCNOT gates without ancilla qubits in $\logarithmic$ depth – up to $70\times$ improvement in gate count** over qubit decompositions. [IEEE Xplore](<https://ieeexplore.ieee.org/document/9069177/>) In 2025, counterdiabatic quantum computing on qutrits improved solution quality $up to 90\times$ compared to qubits** on combinatorial optimization problems including multiway number partitioning and portfolio optimization. [Nature] (<https://www.nature.com/articles/s41534-025-01070-5>)

The SIC-POVM problem in dimension 3 connects qutrits to deep algebraic number theory. Zauner's conjecture (1999) posits that d^2 equiangular lines exist in $\mathbb{C}^d$ for all d ; [Grokopedia](<https://grokopedia.com/page/SIC-POVM>) in $d = 3$, this requires 9 equiangular lines forming a symmetric informationally complete measurement. Exact solutions are known and connect to Hilbert's 12th problem on explicit class field theory. A January 2026 preprint (Joka, arXiv:2601.13475) claims to resolve Zauner's conjecture for arbitrary dimensions, though this awaits peer review.

The novel number systems landscape extends far beyond balanced ternary. Avizienis' signed-digit representations (1961) achieve carry-free addition for radix $r \geq 3$ by allowing digit values $\{-α, ..., α\}$ with $α > r/2$. [Semantic Scholar](<https://www.semanticscholar.org/paper/Signed-Digit-Numbe-Representations-for-Fast-Avizienis/5d46ca96851fd6fdbdb246868033d700776c4262>) [Citationimpact] (<https://www.citationimpact.com/journal-of-computer-engineering-information-technology/signeddigit-number-representations-for-fast-parallel-arithmetic-171060.html>) Gustafson's **posit arithmetic** (2017) replaces IEEE 754 with tapered precision that concentrates accuracy near 1. [Wikipedia] ([https://en.wikipedia.org/wiki/Unum_\(number_format\)](https://en.wikipedia.org/wiki/Unum_(number_format))) [Next Platform] (<https://www.nextplatform.com/2019/07/08/new-approach-could-sink-floating-point-computation/>) Hunhold's **takum arithmetic** (2024) adds logarithmic significands for simpler multiplication, and the **tekum** variant adapts this specifically for balanced ternary hardware. Residue number systems achieve embarrassingly parallel arithmetic through the Chinese Remainder Theorem, [MDPI](<https://www.mdpi.com/2079-3197/9/2/9>) and MIT's **Residue Hyperdimensional Computing** (2025) unifies residue representations with high-dimensional vector algebras for combinatorial optimization. [MIT Press] (<https://direct.mit.edu/neco/article/37/1/1/125267/Computing-With-Residue-Numbers-in-High-Dimensional>)

Yet the Church-Turing thesis imposes a fundamental limit: computational complexity classes are invariant under change of base. [ACM Digital Library](<https://dl.acm.org/doi/abs/10.1137/S0097539795293639>) Converting between binary and ternary takes polynomial time, so P, NP, and BQP are identical regardless of representation. The advantages of ternary are in **constant factors and practical efficiency**, not asymptotic complexity. The one classical domain where balanced ternary is provably optimal is **Bachet's weight problem** (1612): with a two-pan balance, weights of powers of 3 placed on either pan can measure any integer weight from 1 to $(3^n - 1)/2$ [williams]

(https://web.williams.edu/Mathematics/sjmiller/public_html/105Sp10/addcomments/Hayes_ThirdBase.htm) [Groklopedia](https://groklopedia.com/page/Balanced_ternary) — a factor of ~ 1.5 better than binary powers on one pan.

The honest assessment: a constellation, not a unified theory

The mathematical threads examined here form a **constellation of related ideas** rather than a single coherent framework. The strongest connections form a clear chain: balanced ternary $\rightarrow$ symmetric residue mod 3 $\rightarrow$ fractional part centering shift $\{x\} - 1/3 \rightarrow$ Hermite's identity for $n = 3 \rightarrow$ tripling map and Cantor set $\rightarrow$ 3-adic integers and their fractal geometry. This chain is algebraically tight, with exact identities and clean theorems at every link.

The connections to π are genuine but limited. The proven impossibility of base-3 BBP formulas for π itself is not a failure but a mathematically deep result — it reflects the special role of powers of 2 in polylogarithm identities. That π^2 does have a base-3 BBP formula hints at structure yet to be understood. [MetaFilter] (<https://ask.metafilter.com/360896/Math-formula-that-only-works-in-some-base-systems>) [davidhbailey] (<https://www.davidhbailey.com/dhbpapers/bbp-formulas.pdf>) The qutrit quantum Fourier transform naturally incorporates π into base-3 computation through the phases $e^{2\pi i/3^n}$, providing perhaps the most natural bridge.

The higher-dimensional geometry connects primarily through lattice theory. The Eisenstein integers $\mathbb{Z}[\omega]$ bridge FLT (specifically the $n = 3$ case) to hexagonal geometry, [PRIMES] (<https://www.wstein.org/edu/2010/414/projects/ohana.pdf>) [Wikipedia] (https://en.wikipedia.org/wiki/Eisenstein_integer) while lattice problems under various L_p norms underpin post-quantum cryptography. [ScienceDirect] (<https://www.sciencedirect.com/topics/computer-science/lattice-based-cryptography>) The balanced ternary hypercube $\{-1, 0, 1\}^n$ is a genuine geometric object with computational significance, [UCSB] (https://web.ece.ucsb.edu/~parhami/pubs_folder/parh13-asilo-bin-encoded-symm-ternary.pdf) but it remains one lattice among many rather than a universal structure.

Could a new number system or computing model emerge from synthesizing these ideas? The most plausible candidate is not a single system but a **heterogeneous architecture** combining balanced ternary arithmetic (for its algebraic elegance and carry reduction), [Louis-dr] (<https://louis-dr.github.io/ternalu3.html>) qutrit quantum processing (for problems where the 3-level structure provides exponential advantage), and tapered-precision ternary formats like tekum (for floating-point efficiency). The mathematical foundations are solid: radix economy favors base 3, [Williams College +3] (https://web.williams.edu/Mathematics/sjmiller/public_html/105Sp10/addcomments/Hayes_ThirdBase.htm) the Eisenstein integers provide the right algebraic number theory, fractional part identities give exact digit-extraction machinery, and qutrit gate decompositions offer concrete computational speedups. [IEEE Xplore] (<https://ieeexplore.ieee.org/document/9069177/>)

What would be needed to turn this constellation into a framework is a **unifying theorem** — something analogous to how the modularity theorem unified elliptic curves and modular forms to prove FLT. [Wikipedia] (https://en.wikipedia.org/wiki/Wiles's_proof_of_Fermat's_Last_Theorem) One speculative direction: if Hypothesis A could be proved, connecting BBP-type formulas to normality, [Rug] (https://fse.studenttheses.ub.rug.nl/8432/1/Jurjen_Heegen_WB_2008.pdf) and if a base-3 BBP formula for some natural function of π could be found (beyond π^2), this might reveal structural reasons why ternary arithmetic has a privileged relationship to fundamental constants. But as of 2026, no such theorem exists, and

intellectual honesty requires stating clearly that these domains, while richly interconnected, remain fundamentally distinct branches of mathematics that share tools and occasional theorems rather than a common foundation.